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# MUSICRS: BENCHMARKING AUDIO-CENTRIC CONVERSATIONAL RECOMMENDATION

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## ABSTRACT

Conversational recommendation has advanced rapidly with large language models (LLMs), yet music remains a uniquely challenging domain in which effective recommendations require reasoning over audio content beyond what text or metadata can capture. We present **MusiCRS**, the first benchmark for audio-centric conversational recommendation that links *authentic* user conversations from Reddit with corresponding tracks. MusiCRS includes 477 high-quality conversations spanning diverse genres (classical, hip-hop, electronic, metal, pop, indie, jazz), with 3,589 unique musical entities and audio grounding via YouTube links. MusiCRS supports evaluation under three input modality configurations: audio-only, query-only, and audio+query, allowing systematic comparison of audio-LLMs, retrieval models, and traditional approaches. Our experiments reveal that current systems struggle with cross-modal integration, with optimal performance frequently occurring in single-modality settings rather than multimodal configurations. This highlights fundamental limitations in cross-modal knowledge integration, as models excel at dialogue semantics but struggle when grounding abstract musical concepts in audio. To facilitate progress, we release the MusiCRS dataset[1], evaluation code[2], and comprehensive baselines.

***Index Terms*—** Conversational Recommendation, Music Recommendation, Music Information Retrieval, Multimodal Learning

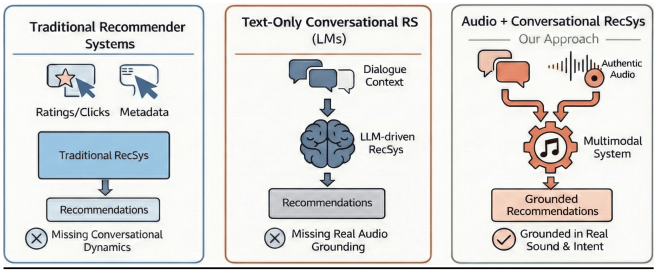
## 1. INTRODUCTION

Traditional music recommender systems rely on collaborative filtering and metadata but ignore the conversational dynamics through which users naturally express preferences. Conversational recommender systems powered by large language models (LLMs) address this limitation, enabling multi-turn interactions that track user intent and produce contextually relevant recommendations. While substantial progress has been made in domains like movies and e-commerce [6, 7, 8], music recommendation presents unique challenges. Listener preferences depend not only on dialogue context but also on musical characteristics such as rhythm, timbre, instrumentation, and production style. These subtle musical dimensions are difficult to capture through text or metadata alone [9], making music an ideal testbed for multimodal conversational recommendation. Current benchmarks predominantly focus on movies or general domains [10], while music datasets emphasize metadata, tags, or listening logs [5] rather than conversations referencing actual recordings. This gap prevents the evaluation of whether systems can bridge natural conversational cues with the musical characteristics driving human preference, limiting progress in multimodal music understanding.

⋆ Equal contribution.

[1] https://huggingface.co/datasets/rohan2810/MusiCRS

[2] https://github.com/rohan2810/musiCRS




| Dataset | Source Type | Real | Audio | Ground. | Rec. | Multi. |
|---|---|---|---|---|---|---|
| CPCD (Playlist) [1] | Crowdsourced | ✓ | ✗ | ✓ | ✓ | ✗ |
| TalkTheWalk [2] | Synthetic (LLM+lists) | ✗ | ✗ | ✓ | ✓ | ✗ |
| LP-MusicDialog [3] | Synthetic (LLM-gen) | ✗ | ✗ | ✓ | ✓ | ✗ |
| Audio Dialogues [4] | Synthetic (captions) | ✗ | ✓ | ✓ | ✗ | ✓ |
| LastFM [5] | User logs | ✗ | ✗ | ✗ | ✓ | ✗ |
| **MusiCRS** | In-the-wild (Reddit) | ✓ | ✓ | ✓ | ✓ | ✓ |

**Fig. 1**: Limitations of existing conversational recommendation approaches (top) and comparison of music recommendation datasets (bottom). MusiCRS is the only benchmark combining authentic conversations, audio grounding, ground truth annotations, recommendation evaluation, and multimodal capabilities.

To address this gap, we introduce **MusiCRS**, the first benchmark for audio-centric conversational recommendation that links conversations with corresponding audio tracks. Built from large-scale Reddit discussions where users naturally exchange music suggestions, the dataset grounds each query in a referenced track, reflecting real-world music discovery. As illustrated in Fig. 1, existing approaches face complementary limitations: traditional systems ignore conversational dynamics, while text-only conversational systems miss real audio grounding. MusiCRS addresses both by integrating dialogue context with authentic audio within a single benchmark. MusiCRS contains 477 high-quality Reddit discussions spanning seven genres, with each conversation grounded in audio through validated YouTube links. This design enables rigorous evaluation of cross-modal understanding in conversational music recommendation.

MusiCRS enables evaluation across three input modality configurations that systematically assess the contribution of different information sources. The *audio-only* configuration tests models using only the referenced audio tracks, isolating the impact of musical content. The *query-only* configuration evaluates models relying solely on conversational text, aligned with traditional text-based recommendation systems. The *audio+query* configuration combines both modalities, testing recent advances in audio-LLMs [11, 12, 13] and multimodal retrieval models [14, 15]. This design enables systematic analysis of how different information sources contribute to music recommendation quality across generative, retrieval-based, and traditional approaches. Our contributions are threefold: